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Forensic Anthropology

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Meaning

- Forensic Anthropology deals with application of anthropological and medical knowledge to questions of law and detection of crime
- "Forensis" – Latin – "Court of Law"
- C.C. Show 1972 – A forensic Anthropologist is ***"a person with specialized knowledge of human sexual, racial, age and individual variations to the problems of medical jurisprudence"***

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Scope

- **Chaurasia**
  1. Whether the bones are human or nonhuman
  2. Whether they belong to one or many humans
  3. What would be the sex
  4. What is the age
  5. What is the race of the person
  6. What is the cause of death
  7. What is the time of death
  8. What other distinguishing characters that lead to personal identification

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Evidences

- Bone, teeth, personal belongs
- Making use of DNA Fingerprinting
- Trivial proportions of saliva, semen, blood or scratches of skin tissue

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Relevance of Anthropology in Forensics

- Experience and Specialization in
  - Comparative Anatomy
  - Racial Affiliations
  - Construction of statures from broken bones
  - Excavation techniques and mapping
  - Interpretive analysis and logicodeductive reasoning

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Identification of Sex

- Sexual dimorphism in pelvic region

Pelvic Girdle	**Typical Male**	**Typical Female**
Sub Pubic Angle	< 90 Degrees	> 90 Degrees
Pubic Shape	Triangular	Rectangular
Sub Pubic Angle Shape	Convex	Concave
Greater Sciatic Notch	< 68 Degrees	> 68 Degrees
Sacrum	Small & Curved	Large & Straight

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FIGURE B-1 Male pelvic girdle.

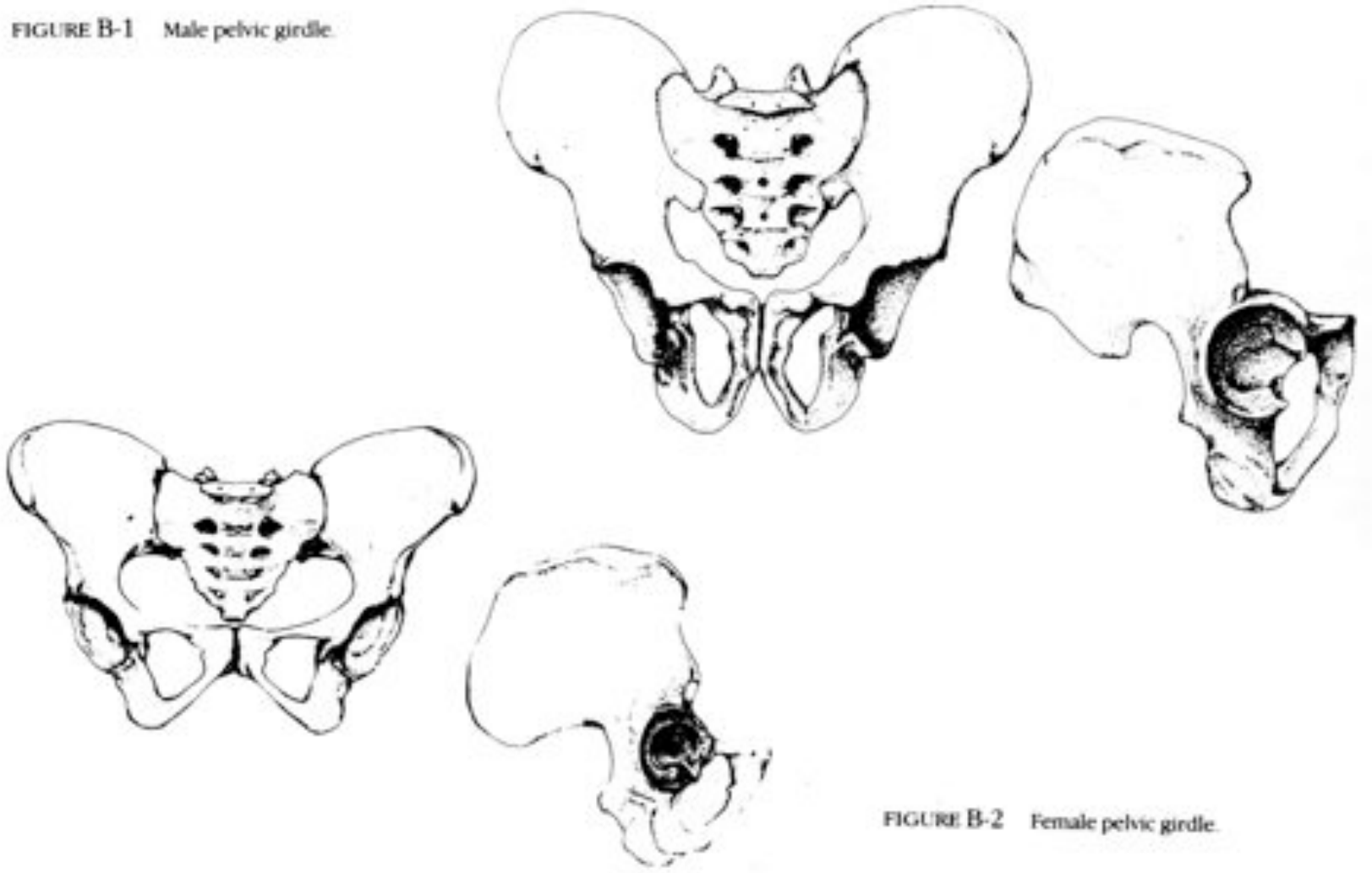


FIGURE B-2 Female pelvic girdle.

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Identification of Sex

- Males are usually larger and have more rugged areas for muscle attachments than do females of the same species
- However, populations differ in skeletal size and robusticity - Asian Indians are smaller and more gracile than Australian aborigines

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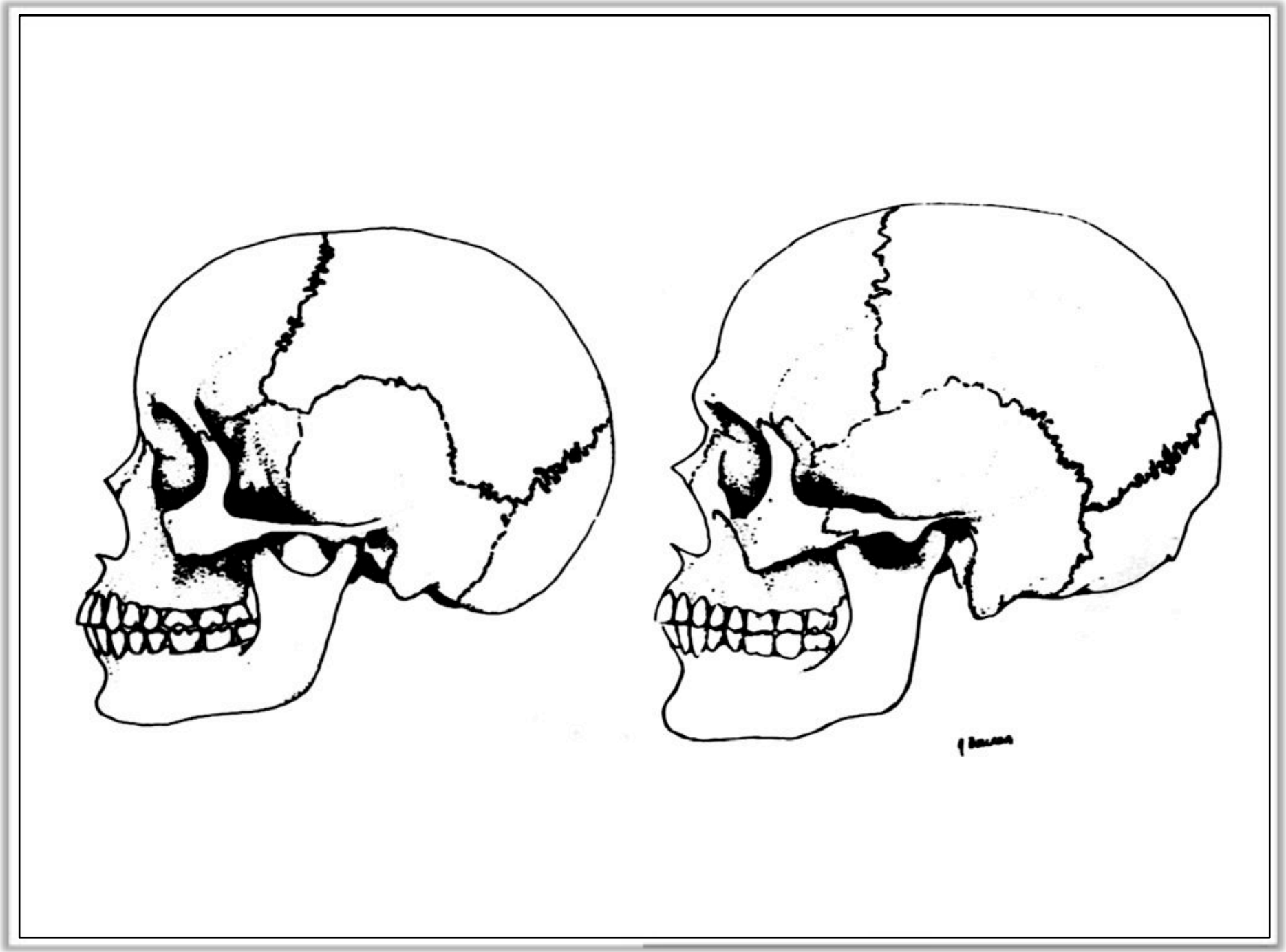
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Identification of Sex

Cranium	Typical Male	Typical Female
Muscle Attachment Areas	More Pronounced	Less Pronounced
Supra Orbital Torus (Ridge)	More Pronounced	Less Pronounced
Frontal Bone	Slanting	Globular
Supra Orbital Rim	Rounded	Sharp
Palate	Deep	Shallow

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Determination of Age

- During growth, the skeleton and dentition undergo regular changes that allow the determination of age at death

Dental Eruption

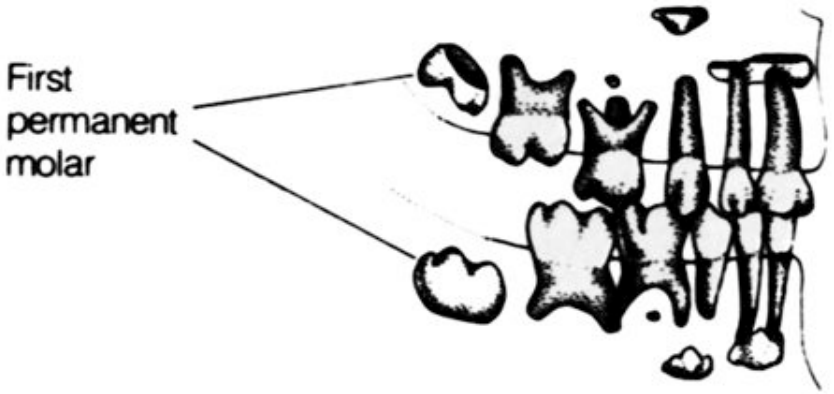
- The determination of ages at which the deciduous and permanent dentition erupts is useful in identifying age to approximately 15 years
- The third molar (wisdom tooth) erupts after this time, but is so variable in age of eruption (if it erupts at all) that it is not a very reliable age indicator

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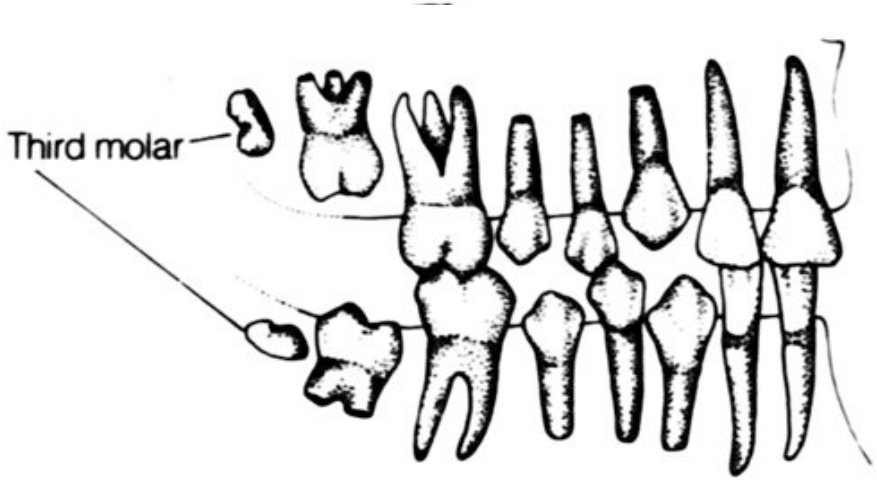
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(*a*) Birth: The crowns for all the deciuuous (milk) teeth (shown in color) are present; no roots, however, have yet formed.



(*b*) 2 years: All deciduous teeth (shown in color) are erupted; the first permanent molar and permanent incisors have crowns (unerupted) formed, but no roots.



(*c*) 12 years: All permanent teeth are erupted except the third molar (wisdom tooth).

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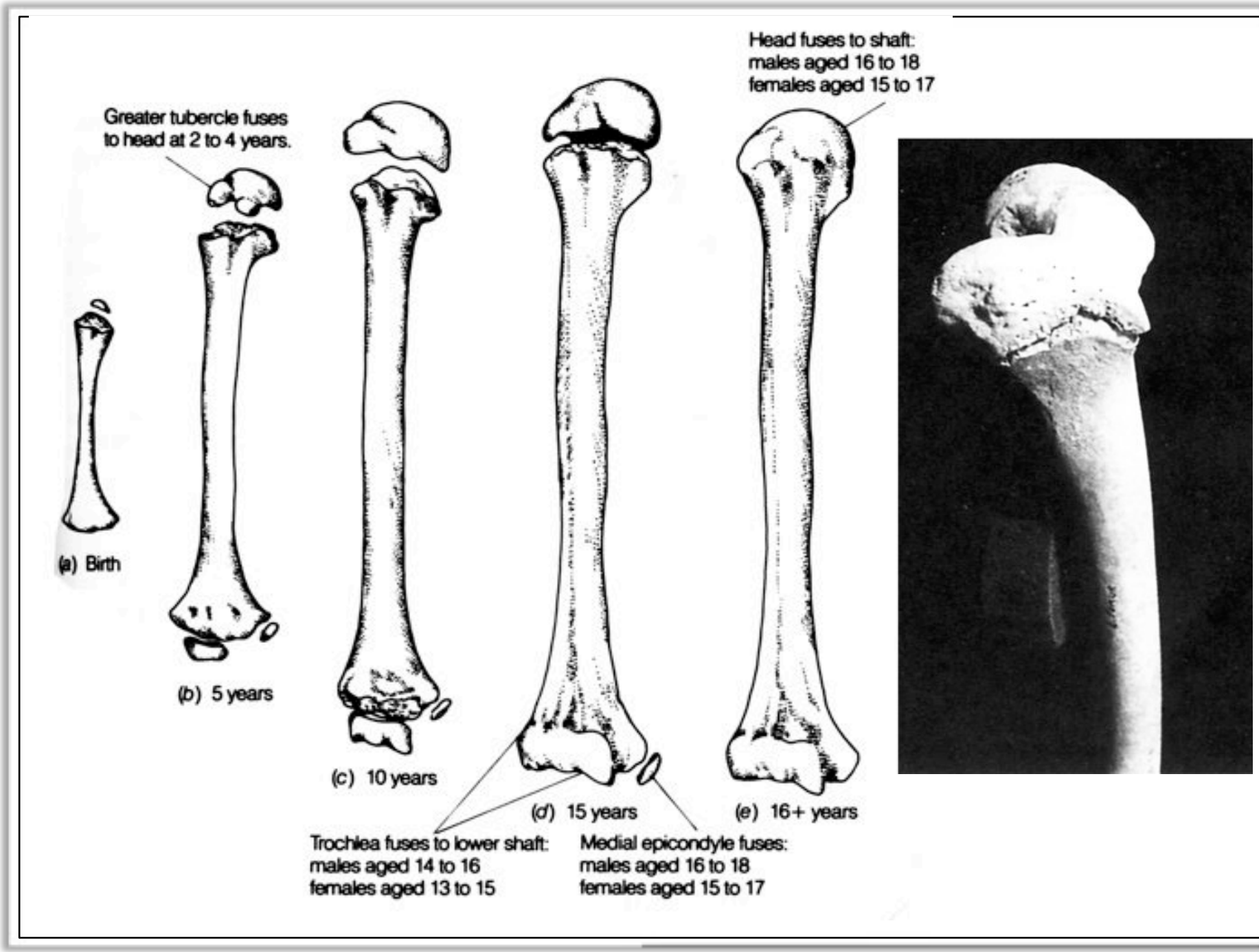
Determination of Age

Bone Growth

- Postcranial bones are preceded by a cartilage model that is gradually replaced by bone, both in the primary growth centers (diaphyses) and in the secondary centers (the ends of the bones, or epiphyses)
- The initial ossification centers are very small, and are only rarely encountered by forensic anthropologist
- The bone continues to grow until the epiphyses fuse with diaphyses
- Because this fusion occurs at different times in different bones, the age of an individual can be determined by which epiphyses have fused and which have not

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Determination of Age

- Females mature more quickly than males, so usually 1 to 2 years must be subtracted from the ages
- Once a person has reached physiological maturity (by early 20s), the determination of age becomes more difficult
- Degenerative changes, including arthritis and osteoporosis, can aid in the determination of relative age, but should not be used by themselves, as injury and certain diseases can cause changes that mimic old age in bones

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Determination of Race

FEATURE	NEGROID	CAUCASOID	MONGOLOID
Incisors	Blade	Rarely Shoveled	Shoveled
Cranial Shape	Dolicocranic	Mesocranic	Brachycranic
Nasal Root	Wide, Rounded	Narrow, Pinched	Medium, Tented
Nasal Aperture	Platyrrhiny	Leptorrhiny	Mesorrhiny
Zygomatic Bone	Medium	Retreating	Projecting
Ear Opening	Round	Round	Oval
Facial Shape	Prognathic	Orthognathic	Medium

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Stature

- Stature is usually calculated by measuring the length of long bones by Trotter, Gleser and Genoves
- Because these formulae are derived for each sex of various racial groups, the sex and race of the unknown individual must be known before formulae can be reliably applied

Femur Length (mm)	**Stature (cm)**
452	169
456	170
461	171
465	172
469	173

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Photo superimposition

- Photo superimposition is employed when a probable identification has been made
- The skull is superimposed onto a photograph of the known individual
- If enough landmarks fall on the same position on the skull **and** on the photograph, the forensic scientist has made a identification
- This technique is also used to reinforce the decision made in some of the cases

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Facial Reconstruction

- Facial reconstruction, also termed facial reproduction is a process used when other identification procedures have been unsuccessful
- Two different methods of producing a face on the skull are employed
  - A portrait of the individual using clues from the bones of the face
  - A more direct, 3D method of applying clay to the skull
- These techniques employ both science and art

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Facial Reconstruction

- The physical anthropologist discovers the age, sex and race of the skull, but there is no direct evidence from bone that indicates the eye color, hair color and style, lip form, or degree of wrinkling in the individual
- There is a great deal of subjectivity in the rendering of the finished product,
- An exact reproduction, is however, not expected, only a likeness

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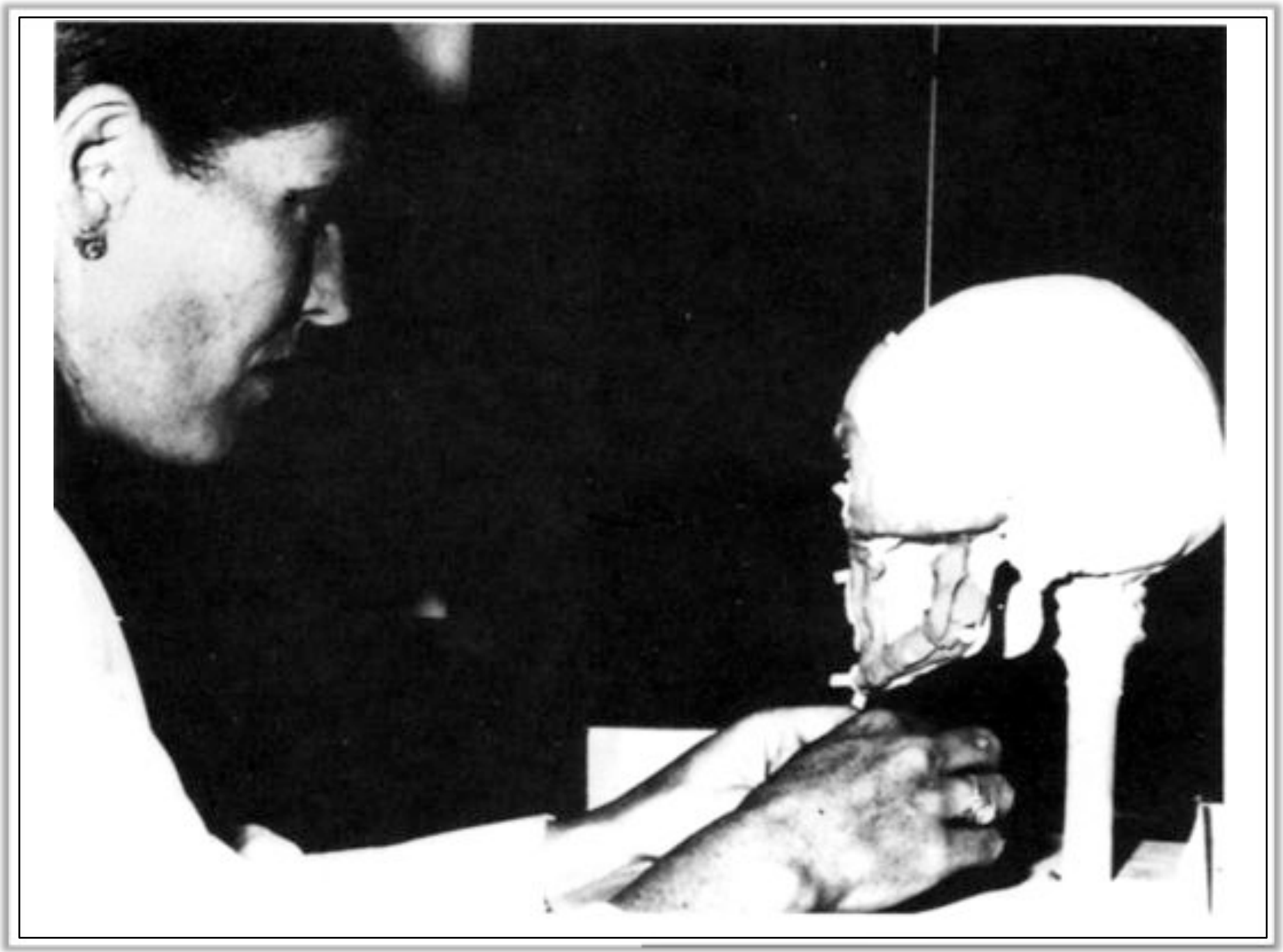
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Facial Reconstruction

- **Erasers** or blocks of clay marking tissue depths are commonly glued to the skull (depths are arrived at experimentally from cadavers)
- Clay strips, graduated to the various tissue depths, then "fill in the dots" between erasers and the face is "fleshed out"
- The eyes, nose and sometimes ears are then fashioned according to various guidelines, and a wig is usually added
- If required, features for aging are also added

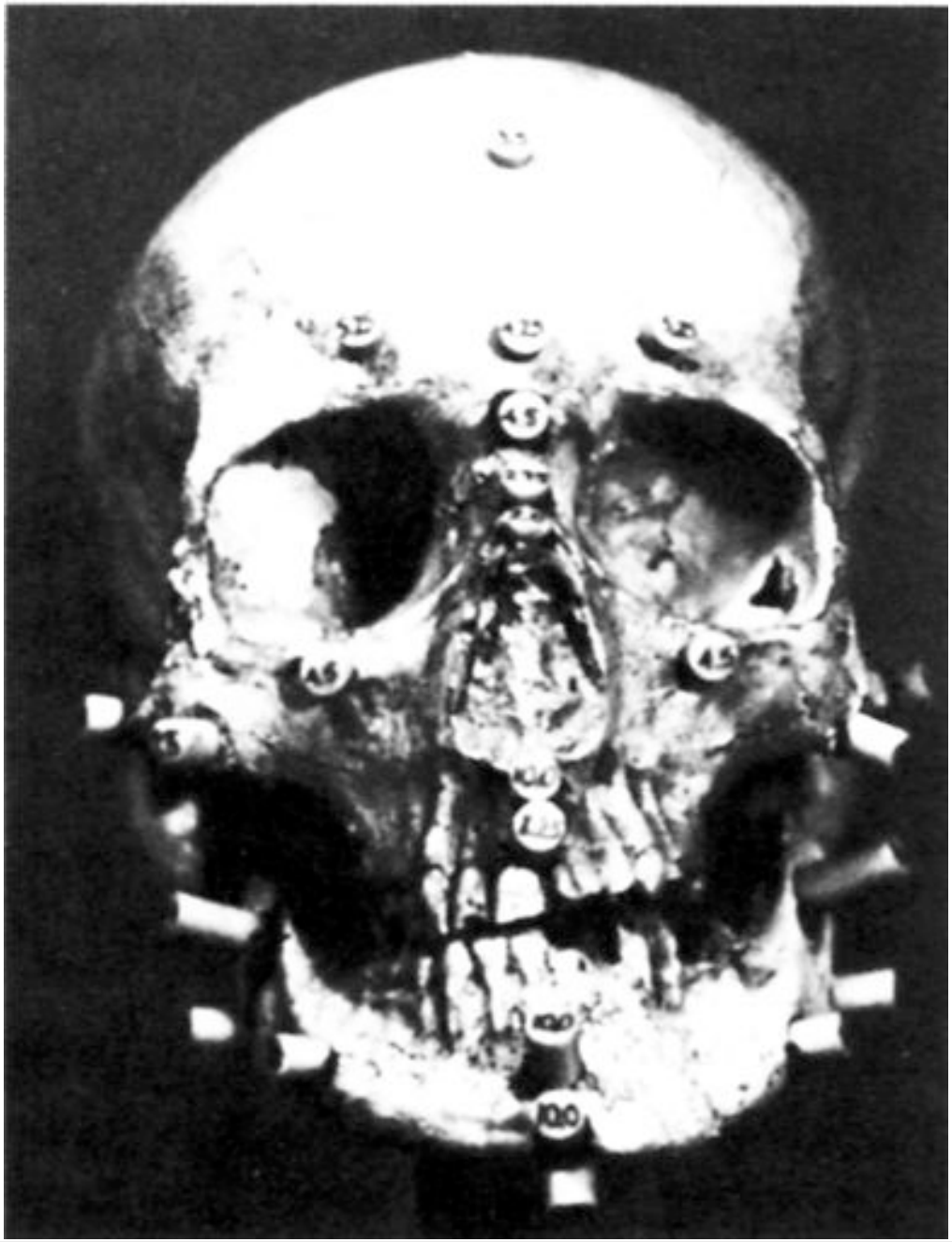
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FIGURE e and f A nose and lips are added and refined.

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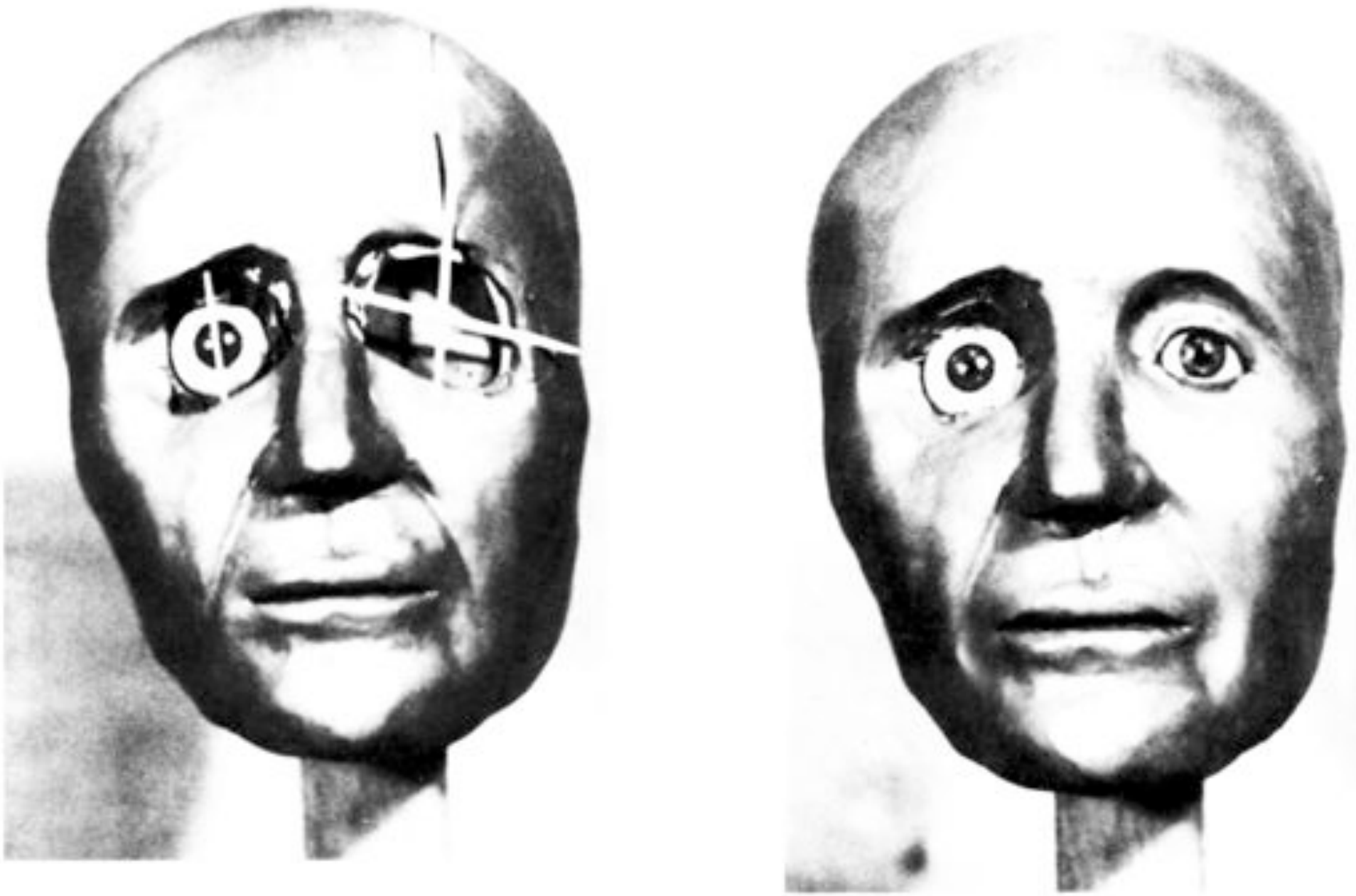


FIGURE g and h Glass (or plastic) eyes are placed into orbits and eyelids are fashioned.

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FIGURE i and j Completed reproduction. (i) Before adding aging features. (j) After "aging" the face to correspond with aging indicators ascertained from other parts of the skeleton.

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Kinanthropometry

- Though sports have evolved as a part of human culture, the biological aspects of man like body size, body proportions, physique, nutrition - continuously influence the performance in sports
- Most of these features are controlled by both heredity and environment
- Apart from this, human psychological factors like training, motivation and nature also influence the performance
- All this proves that it requires an interdisciplinary approach combining human biological, cultural and psychological aspects to understand the environment of sports

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Kinanthropometry

- This constituted the foundation of anthropology in study of sports
- The study of sports in anthropology constitutes a specialized sub discipline – Kinanthropometry
- This branch evaluates physical structure of individuals in relation to gross motor functions
- It includes the study of such aspects like maturation, nutrition and body composition

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History

- The term "Kinanthropometry" was coined by Bill Ross – 1972
- It was first considered in Olympic Scientific Congress at Quebec in 1976, prior to Montreal Olympics in the same year
- The driving force behind most initiatives for development of Kinanthropometry has been **UNESCO** when it founded an **International Working Group on Kinanthropometry** at Brasilia, Brazil
- This group has been working under **International Council of Sports Science and Physical Education,** a division of UNESCO

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Factors in Sports

- The factors that influence the performance in sports are controlled by heredity and environment
- Genetics, however, has a greater role to play in the formation of phenotypes like size, physique, body composition, metabolic powers, strength, speed and skill, cardiovascular adaptations etc.
- Environment can shape a genotype into a fit type by way of training and motivation
- Kinanthropometry aims at selecting the fit genotypes which help individuals attain their fullest potentialities

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Factors in Sports

- Apart from muscle strength, a sport will also depend upon coordinated body movements
- Anthropometrical studies of individuals can help select players who can have better potentialities in a particular sport than others
- Training and other extragenic influences can only change one's phenotype within the limits set by the genotype
- It is thus imperative to lay more emphasis on the genetically determined morpho-physiological status of an individual

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Somatotypes

- By making use of the techniques of Kinanthropometry, anthropologists have been able to classify humans into different somatotypes and suggest the correct sports for them
- **Koherausch – 1929**
  - Slender Type – Runners, Jumpers and Hurdlers
  - Medium Type – Boxers, Ball Players, Swimmers
  - Massive Type – Weight Lifters, Throwers & Wrestlers

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Somatotypes

- **Arnold – 1939**
  - Gymnast Type – Relatively long hulk coupled with breadth
  - Wrestler Type – Mighty forms, great breadth of shoulders and pelvis coupled with greater depth and breadth of shoulders
  - Pentathlon Type – Individuals with medium to slender body types with relatively longer legs and less body breadth
- Apart from this, body composition including muscular, skeletal tissues and fatty tissues have to be considered along with motivational forces, training and psyche
- Anthropology also helps in designing sports equipment

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